

Overview

This report uses Tableau to analyze the client's investment portfolio in order to assess risk, performance, and asset allocation. The goal is to determine how each asset class adds to the overall value of the portfolio and find ways to increase stability and diversification.

Portfolio valuations, stocks and ETFs, main and minor asset classes, and performance from 2019 to 2022 are all covered by the dataset. To illustrate portfolio composition, concentration, and the correlation between risk and return, several visualizations were created.

Chart by Chart Analysis

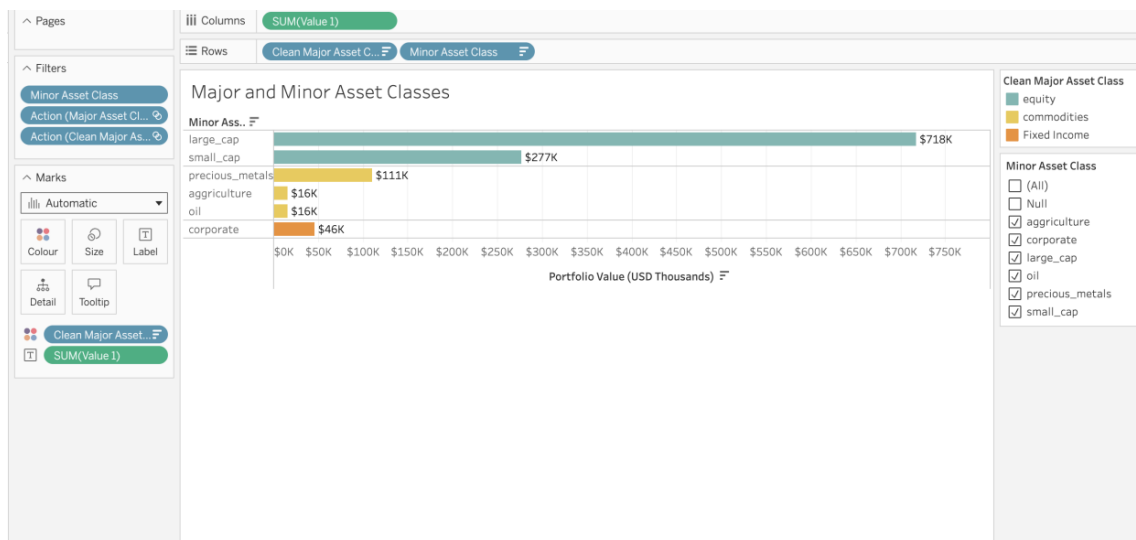
1. Major and Minor Asset Classes (Portfolio Composition)

Observation:

With large cap assets valued at about \$718K and small cap assets at about \$277K, equity holdings make up the majority of the portfolio. Smaller shares are contributed by commodities, fixed income, and other categories, which together account for less than one-third of the overall value.

Insight:

This composition emphasizes an equity focused, growth-oriented allocation. This structure exposes the client to more volatility even if it can increase returns in favorable conditions. Increasing diversification into non-equity asset classes would lessen reliance on market cycles and increase portfolio resilience.



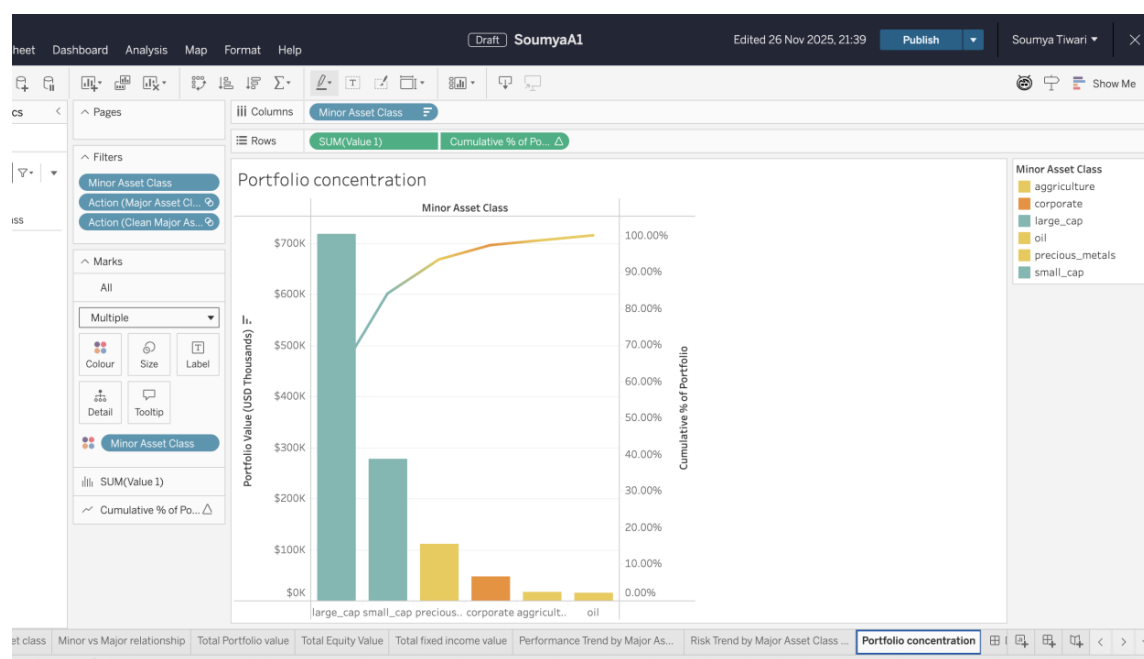
2. Portfolio Concentration (Pareto Analysis)

Observation:

The Pareto chart shows that over 70% of the total portfolio value is accounted for by the top two asset classes, large-cap and small-cap, with the cumulative curve flattening drastically beyond these holdings.

Insight:

Due to the portfolio's high concentration, risk is not evenly spread. Concentration raises susceptibility to sector-specific downturns even while it can enhance benefits. Exposure could be balanced by adding new holdings in under-represented sectors like commodities or fixed income.



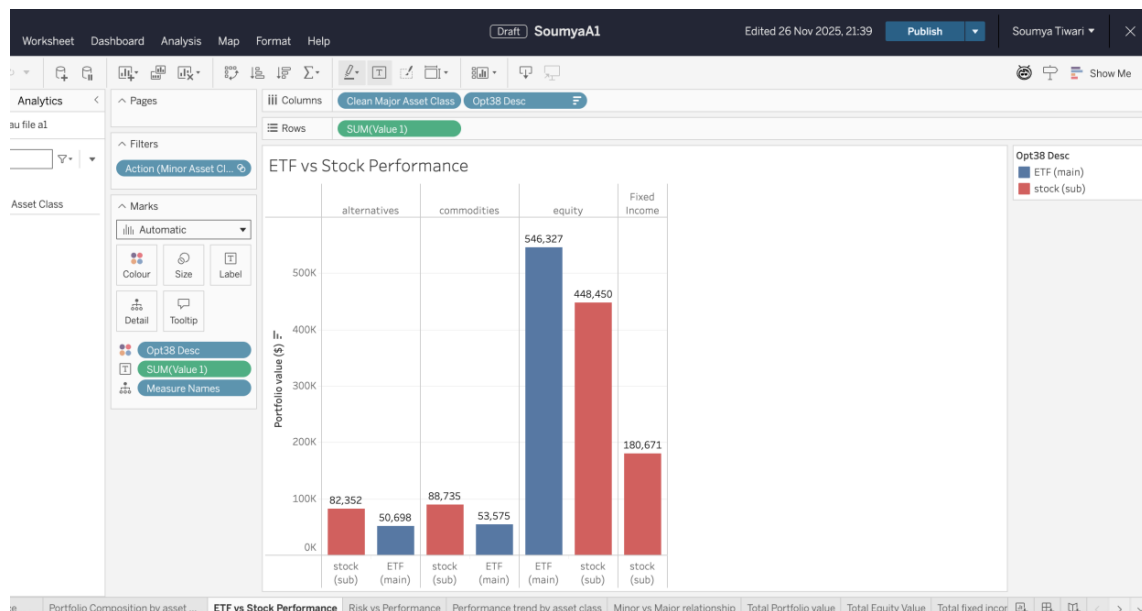
3. ETF vs Stock Performance

Observation:

ETFs perform better than individual stocks in terms of overall value and consistency across all main classes. While commodities and fixed income continue to perform modestly and steadily, equity ETFs exhibit greater growth and less volatility than their stock equivalents.

Insight:

ETF allocations lower unsystematic risk associated with individual holdings and offer effective diversification. Increasing ETF exposure can increase overall portfolio stability without compromising performance potential, particularly in high-volatility categories.



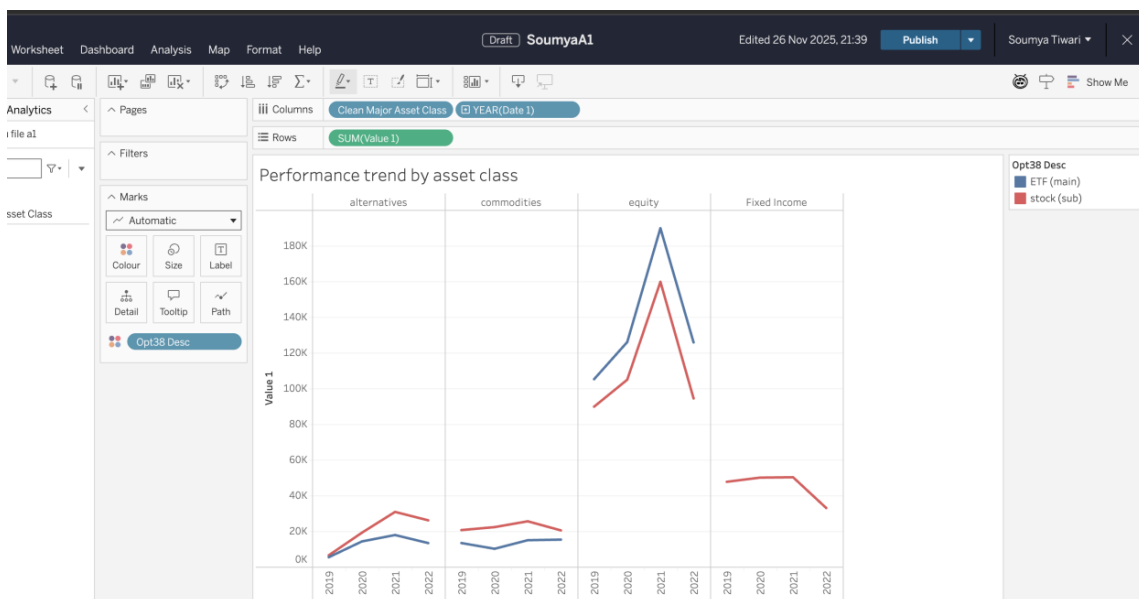
4. Performance Trend by Major Asset Class (2019–2022)

Observation:

Performance rose gradually starting in 2019 and peaked in 2021 before marginally falling in 2022. Commodities and fixed income performed comparatively steadily, but stocks had the biggest volatility.

Insight:

The portfolio's vulnerability to cycles in the equity market is confirmed by the 2021 performance spike, which coincides with times of high market optimism. When equity performance declines, stable returns from non-equity classes imply that diversification reduces negative fluctuations.



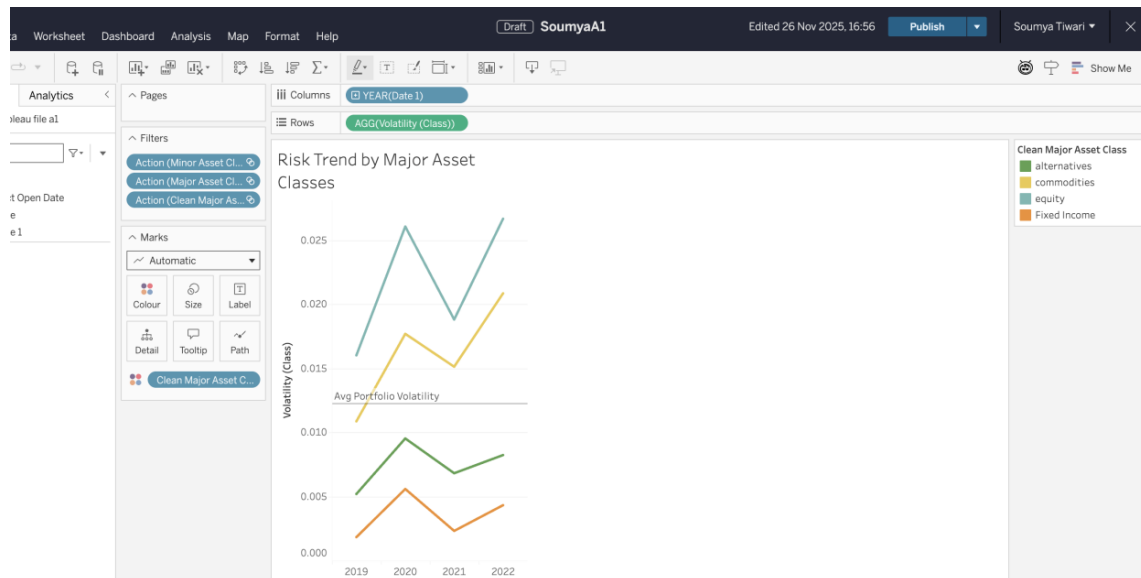
5. Risk Trend by Major Asset Class

Observation:

The trend of volatility was similar, with a strong increase in 2021 and a stabilization by 2022. The average portfolio volatility is indicated by the interactive reference line, which dynamically recalculates when filters are applied.

Insight:

The majority of portfolio risk is driven by stocks, surpassing the average benchmark in 2021. Commodities and fixed income continuously drop below this threshold, confirming their function as protective investments. Early warning indicators of market stress can be obtained by tracking volatility fluctuations.



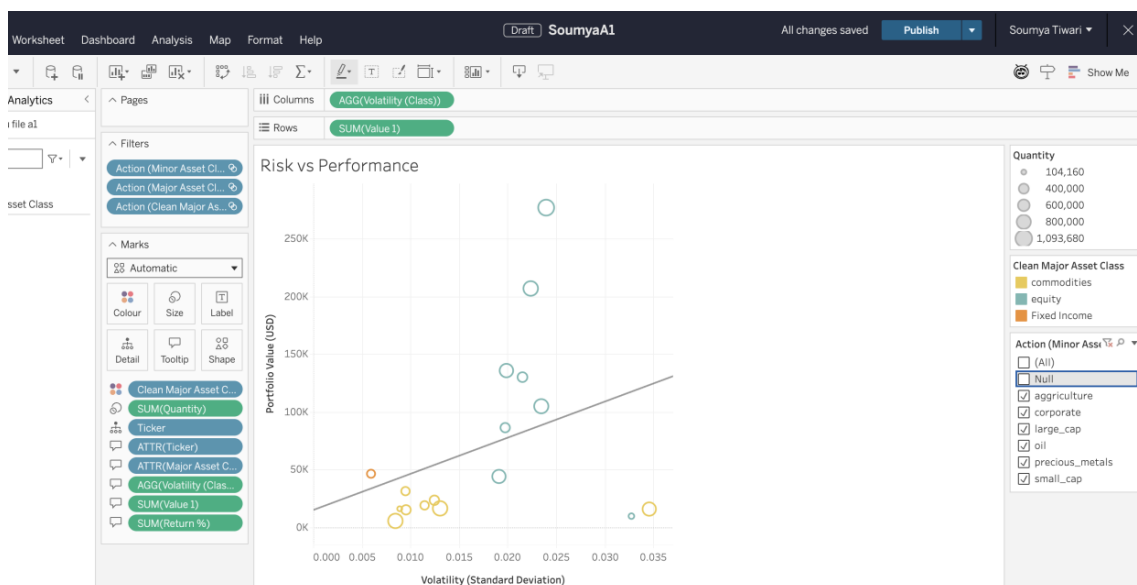
6. Risk vs Performance Scatter Plot

Observation:

There is a definite positive association between more volatility and higher returns, as seen by the scatter plot. Equities are located in the upper-right quadrant, but the majority of points cluster in the centre, indicating balanced returns and modest risk.

Insight:

Although the portfolio maintains a reasonable risk-return trade-off, stocks exhibit the most aggressive profile. A percentage of equity exposure might be shifted to assets with a moderate level of risk, which would improve overall portfolio stability and preserve rewards.



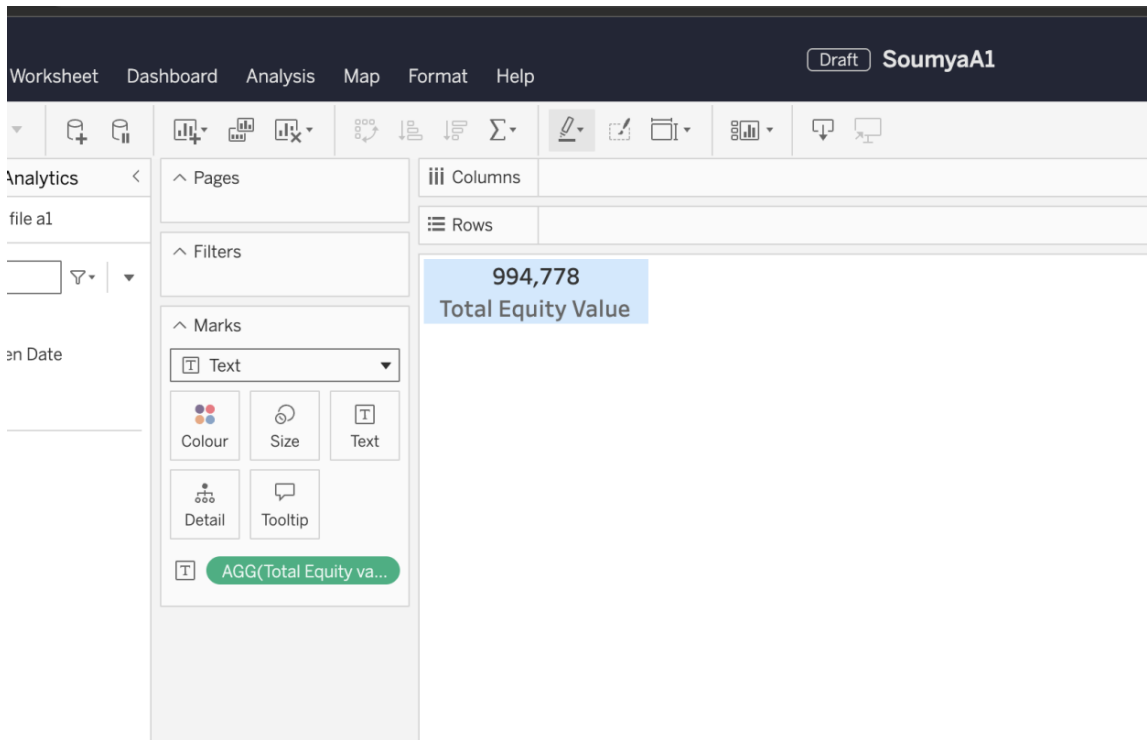
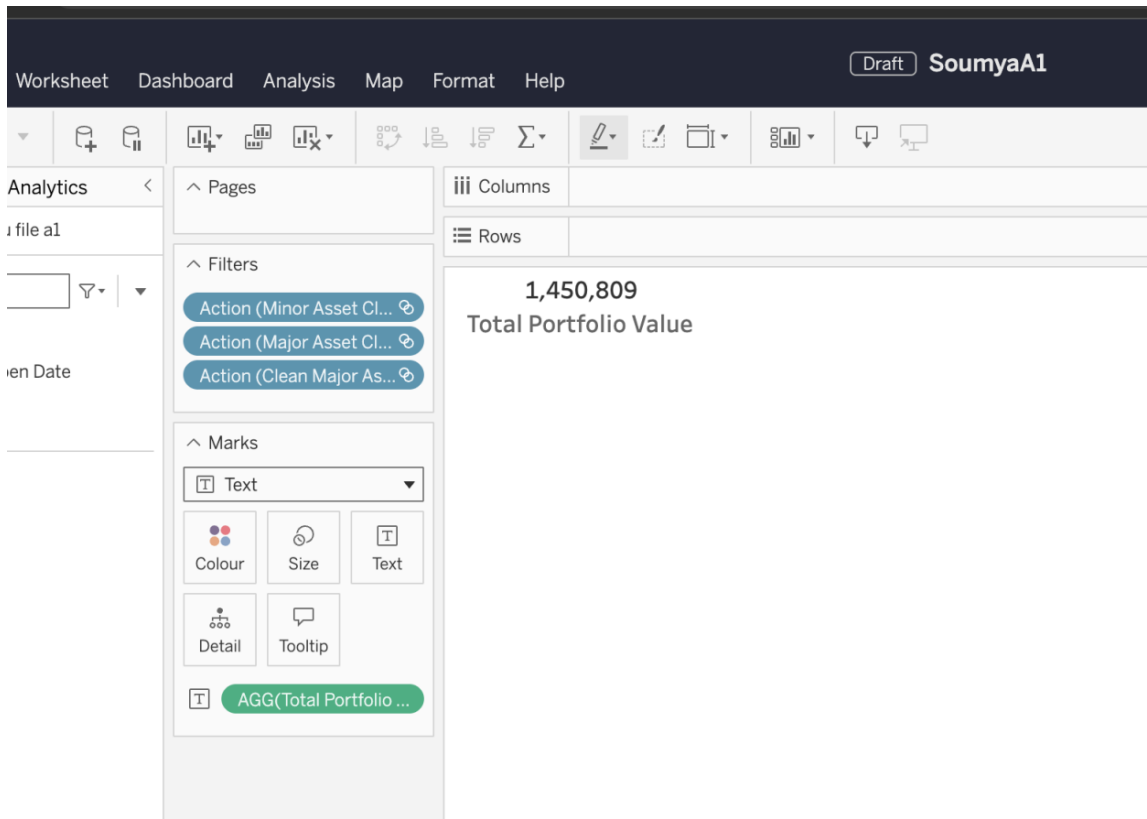
7. KPI Summary

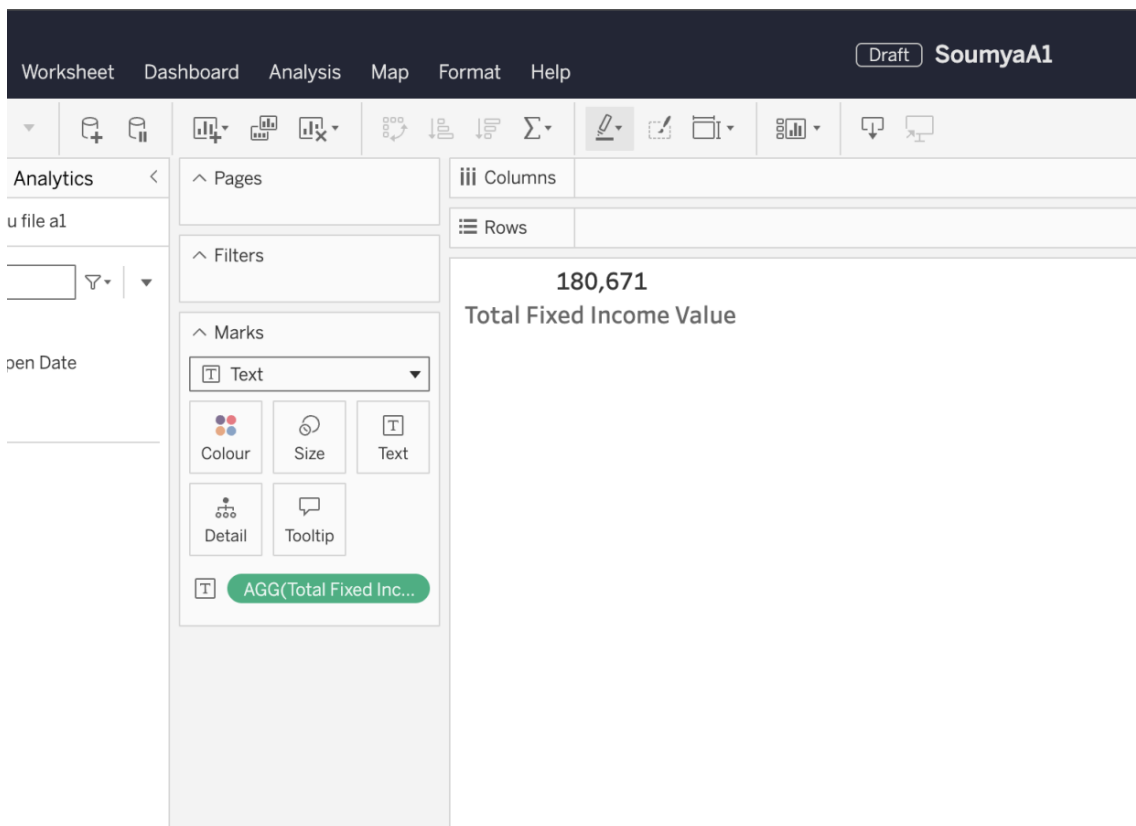
Observation:

According to the key measures, the total portfolio value is \$1.45 million, with \$994K ($\approx 69\%$) coming from stocks and \$181K ($\approx 12\%$) from fixed income.

Insight:

These numbers highlight the importance of wider diversification and the predominance of equities holdings. Increasing fixed-income exposure may improve risk-adjusted performance and lower overall volatility.





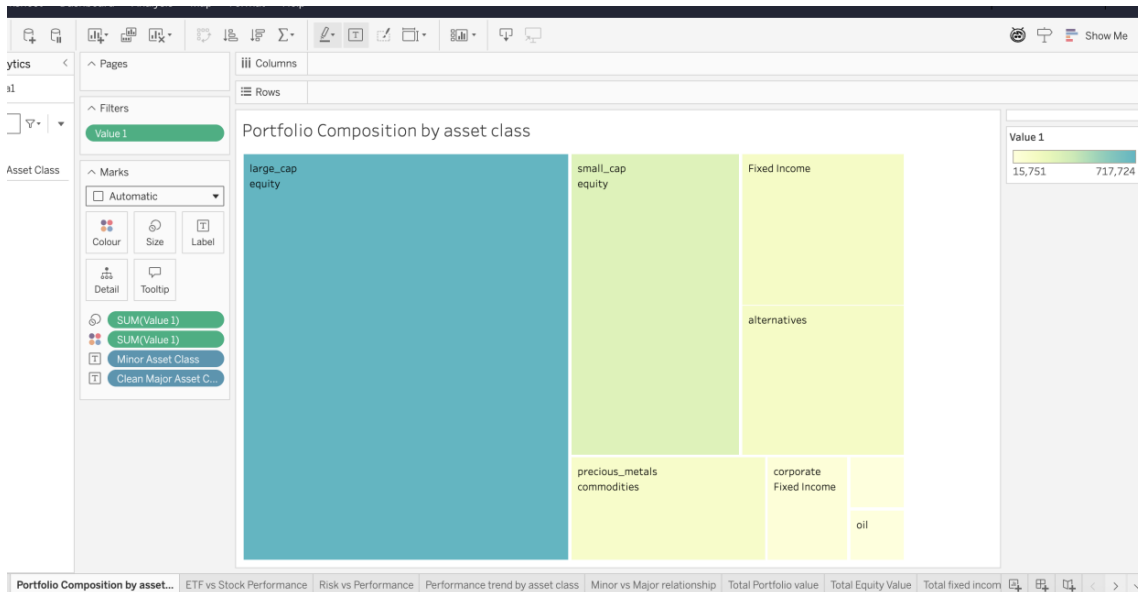
8. Tree Map (Portfolio Overview)

Observation:

The portfolio imbalance is graphically confirmed by the tree map, which shows that other asset classes are minor while large-cap and small-cap stocks occupy the greatest portions.

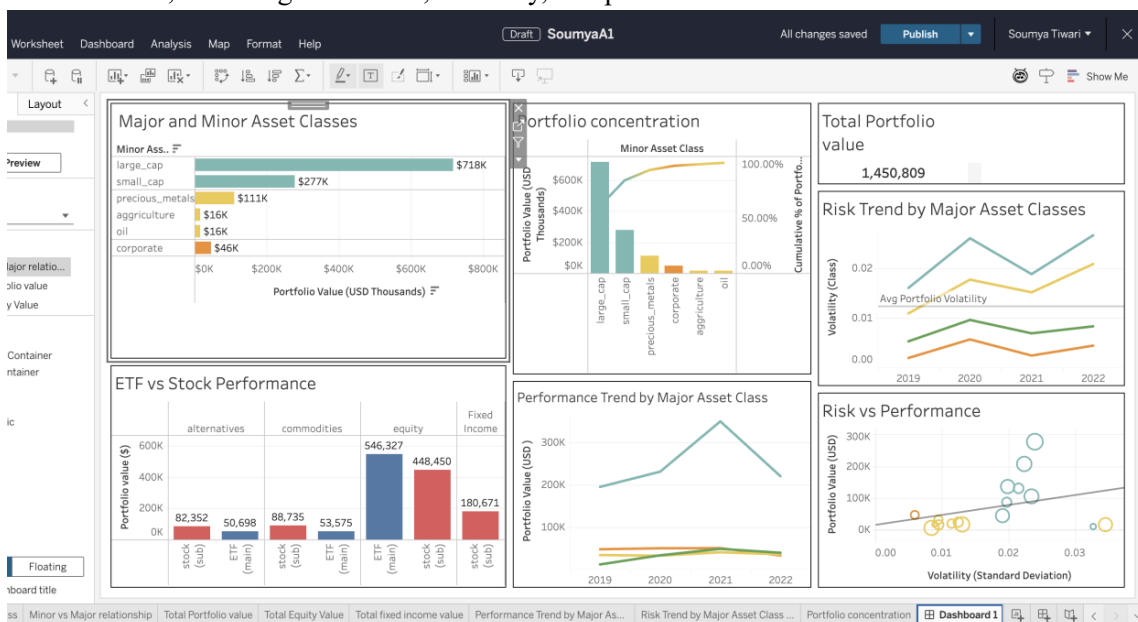
Insight:

The conclusion that the portfolio is still equity-heavy and would benefit from rebalancing is reinforced by this visualization, which concisely summarizes concentration and allocation trends seen across earlier charts.



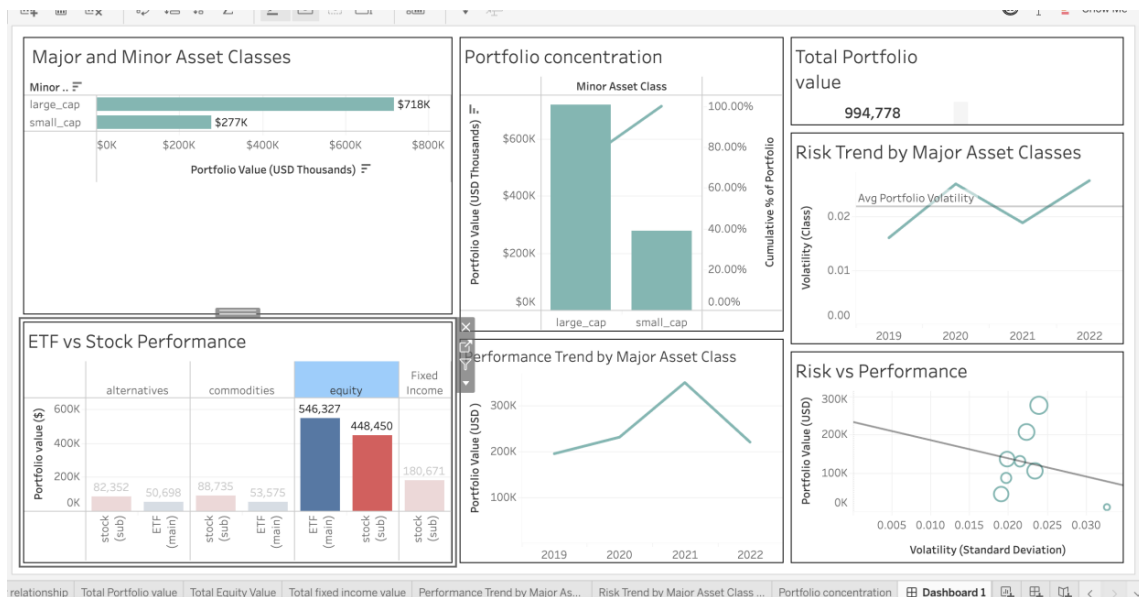
Integrated Dashboard and Interactivity

The final Tableau dashboard allows for real-time analysis of the connections between asset class, performance, and risk by combining all visualizations into a single interactive view. Because each chart is linked by filter actions, choosing any asset class instantaneously updates all the visualizations, including trend lines, volatility, and portfolio value and concentration.



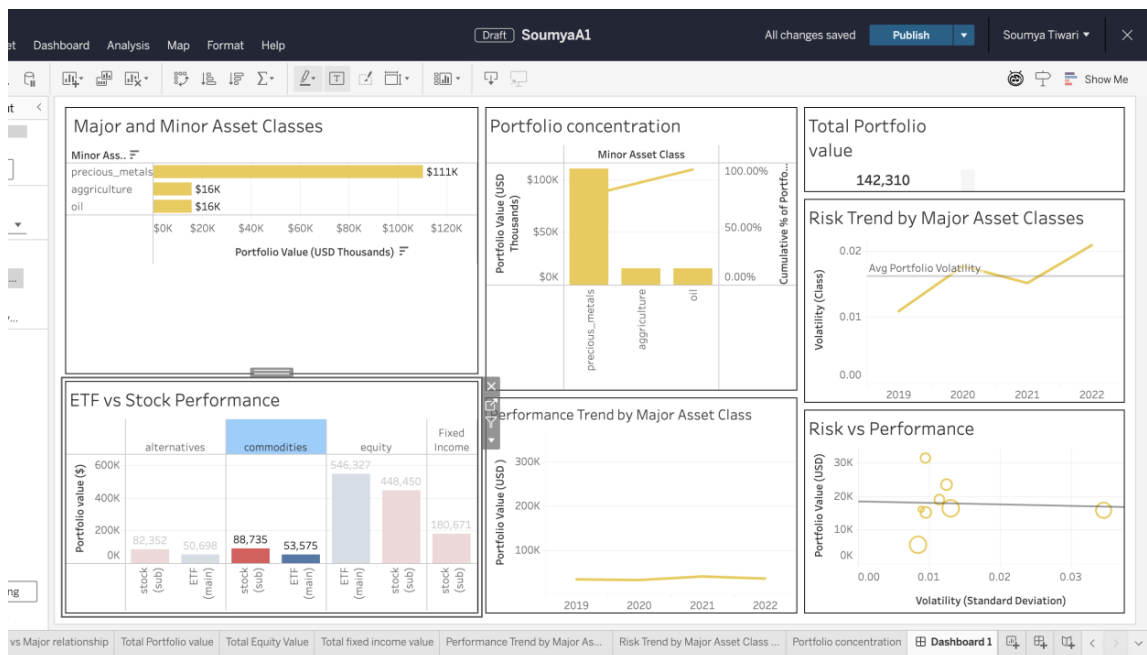
Scenario1

In the first scenario, filtering to Equity shows a total portfolio value of \$994K, while volatility rises above the average benchmark, highlighting the risk-return trade-off from concentrated equity exposure.



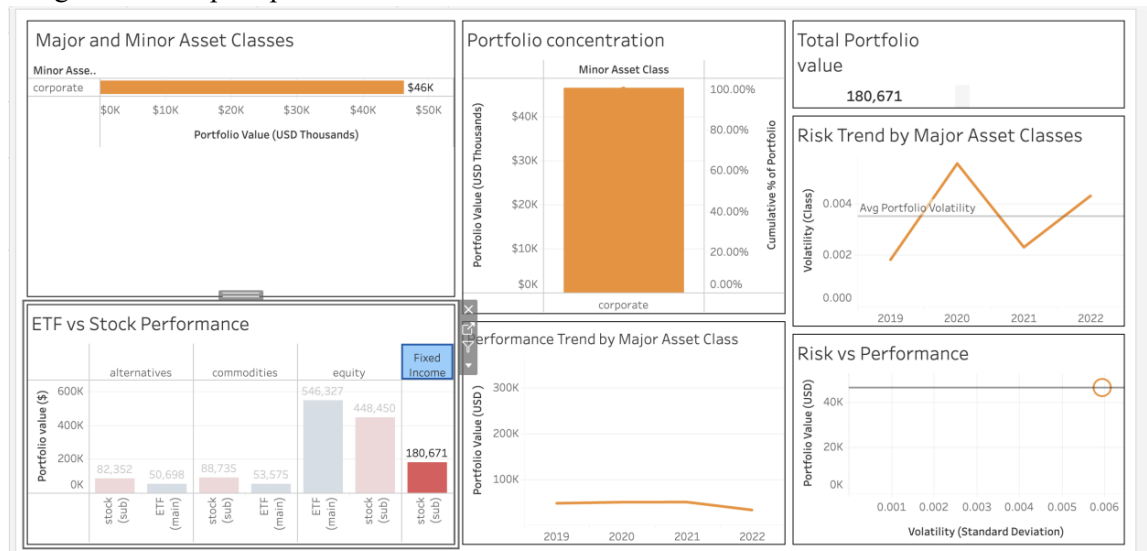
Scenario2

In the second scenario, selecting Commodities reveals lower portfolio value but also significantly reduced volatility, demonstrating the stabilizing role of non-equity assets.



Scenario3

When filtered to *Fixed Income*, the portfolio exhibits the lowest volatility across all asset classes, confirming its defensive nature. Returns are modest but stable, reinforcing its role in risk mitigation and capital preservation.



Findings

Across all asset classes, the portfolio analysis shows three different patterns of behavior. With almost 65% of total holdings (\$994K), equities account for the majority of the portfolio's value and are responsible for both performance and volatility. better volatility is correlated with better returns, according to the Risk vs. Performance chart, which shows an aggressive risk-return profile in line with growth objectives.

The diversification benefit of commodities is demonstrated by their smaller share (\$142K) and lower volatility than the average of the portfolio. The Risk Trend chart validates commodities as stabilizers against stock swings by demonstrating continuous risk control across years despite moderate returns.

With a \$180K portfolio valuation and risk levels well below the mean, fixed income shows the least volatility and the most consistent performance trend. Its defensive role in the portfolio is strengthened by its steady, low-variance profile, which guarantees liquidity and downside protection.

The benefit of diversification is quantitatively demonstrated by interactive filters, which verify that volatility falls, and performance variation narrows when the portfolio moves from equity-heavy to fixed-income or commodity-weighted compositions.

Business Insights

The interactive dashboard shows that the client is vulnerable to systemic market shocks because the portfolio concentration is still biased towards equity. Although stocks provide profits, their volatility may reduce value during downturns, according to the Performance Trend by Major Asset Class and Risk Trend graphics. Fixed income guarantees capital stability, whereas commodities provide countercyclical resilience. When taken as a whole, these results indicate that the client's risk exposure surpasses their ideal tolerance threshold, necessitating a purposeful rebalancing towards balanced diversification.

Additionally, cross-filter interactions provide empirical evidence for multi-asset portfolio

optimization by demonstrating that asset correlations differ structurally fixed income and commodities preserve lower correlation coefficients to stocks.

Recommendations

- **Strategic Rebalancing:** To mitigate volatility without significantly lowering projected returns, reduce equity weighting by 10% to 15% and reallocate to fixed income and commodities.
- **Dynamic Risk Monitoring:** Use filter actions for scenario testing and keep the existing dashboard up to date as a live monitoring tool to track changes in correlation and volatility.
- **Performance Governance:** Conduct evaluations every three months to assess asset-specific As macro conditions change, sharpen ratios and make dynamic adjustments to allocations.
- **Client Advisory Messaging:** Stress sustainable returns above speculative profits by portraying the diversified structure as a disciplined, risk-managed growth plan.

Conclusion

Overall, the portfolio shows a strong growth bias that is tempered by moderate exposure to fixed income and commodities and backed by equity-driven returns. Diversification significantly lowers volatility while maintaining performance, as the interactive dashboard confirms. Nevertheless, the customer is exposed to cyclical risks with the existing allocation, which may jeopardize long-term stability. The client can increase resilience and consistency in returns by carefully rebalancing towards lower-volatility assets, keeping an eye on risk in real time, and adjusting portfolio weightings to reflect shifting market conditions. The dashboard functions as an analytical and decision-support tool, enabling management to test scenarios, visualize performance, and make data-driven allocation decisions that improve the portfolio's performance discipline and long-term sustainability.

